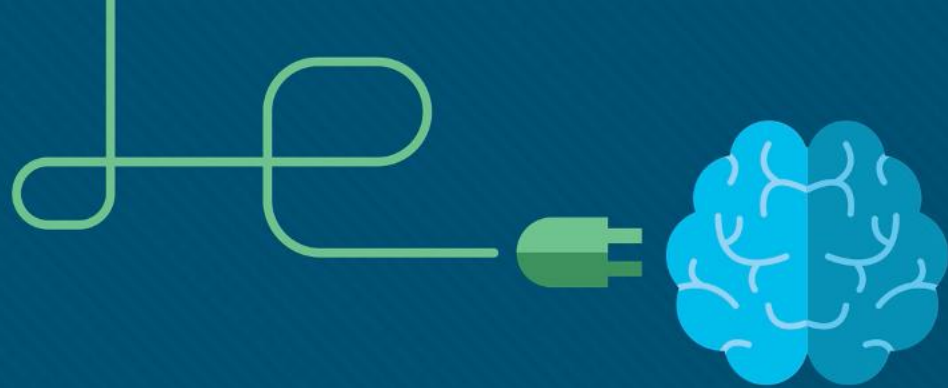




Module 3 Exploring Visual and Interactive Elements

Introduction to HTML



Module Objectives

Module Title: Introduction to HTML

Module Objective: In this module, you'll learn to enhance your web pages with images, hyperlinks, multimedia, and forms. You'll explore the essentials of creating interactive and visually engaging web content, building a solid foundation for user interaction and dynamic design.

Skills you'll gain: Adding images, embedding multimedia, creating hyperlinks, and designing functional HTML forms.

At the end of this module, you'll complete a project that incorporates interactive elements and a test to solidify your understanding.

Module Objectives

Course Syllabus:

The course is organized into six modules, each focusing on key aspects of HTML, along with a final project and test to solidify your learning. Below is a detailed breakdown of the syllabus:

Module 3: Exploring Visual and Interactive Elements

- Images
- Hyperlinks
- Media
- Essentials of HTML Forms
- Forms – Beyond the Basics
- Building HTML Forms
- Module 3 Completion

3.1 Images

3.1.1 The Tag and Its Attributes

Images are an essential element of any website. They not only add visual interest but can also help to communicate information to users. In HTML, you can easily add images to your web page using the tag.

The tag is an empty element, which means that it does not have a closing tag. Instead, all the attributes are included in the opening tag.

The most important attributes of the tag are <src>, <alt>, <width>, and <height>.

The src Attribute

The src attribute specifies the source URL of the image that should be displayed.

For example: src="https://edube.org/uploads/media/default/0001/03/mountain_landscape.jpg".

The alt Attribute

The alt attribute provides alternative text that will be displayed if the image cannot be loaded. It is also used by screen readers for visually impaired users.



For example: alt="Mountain landscape".

3.1.1 The Tag and Its Attributes

The width Attribute

The width attribute specifies the width of the image. It can be defined in pixels (e.g., width="340") or as a percentage of the containing element's width (e.g., width="50%").

The height Attribute

Similarly to width, the height specifies the height of the image in pixels or as a percentage of its parent container.

For example: height="250" or height="50%".

3.1.2 Image Formats

Before inserting an image into your web page, it's essential to choose the right image format. The most common image formats used on the web are:

- JPEG,
- PNG, and
- GIF.

JPEG is used for photographs, while PNG is best suited for graphics and images with transparent backgrounds. GIF is commonly used for simple animations and graphics.

Image Formats in HTML

Format	Description	File Extension
JPEG (Joint Photographic Experts Group)	Best suited for photographs or images with many colors and gradients. JPEG images are compressed, which means that they have a smaller file size, but the compression can result in a loss of quality.	.jpg or .jpeg
PNG (Portable Network Graphics)	Best suited for images with transparent backgrounds or images with sharp edges, such as logos or graphics. PNG images are lossless, which means that they do not lose quality when compressed.	.png
GIF (Graphics Interchange Format)	Best suited for simple animated graphics or images with limited colors, such as icons or buttons. GIF images can also be used to create simple animations.	.gif

3.1.2 Image Formats



PNG – Portable Network Graphics



JPG, JPEG – Joint Photographic Experts Group



GIF – Graphics Interchange Format



SVG – Scalable Vector Graphics

Although JPEG, PNG, and GIF are the most commonly used image formats, keep in mind that there are other image formats available, such as SVG (Scalable Vector Graphics) and WebP.

SVG is a vector-based format that is ideal for images that need to be scaled without losing quality, while WebP is a relatively new format that offers better compression and smaller file sizes than JPEG and PNG. However, not all web browsers support these formats, so it's important to check compatibility before using them on your website.

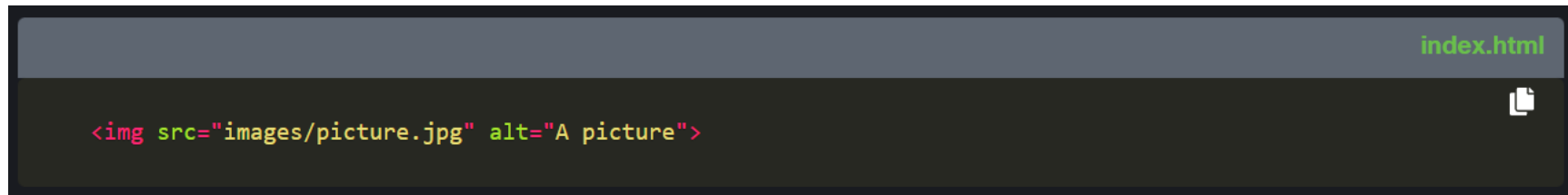
3.1.3 Image Location

When adding images to your website, it's important to consider the location of the image file. If the image is stored on your computer, you will need to upload it to your website's server. If the image file is not in the same directory as the HTML file, you'll need to provide a relative or absolute path to the image file.

Once the image is in the right location, you can use the `<img>` tag and the `src` attribute to insert it into your web page.

Relative Path

A relative path is a path that starts from the directory of the HTML file. For example:



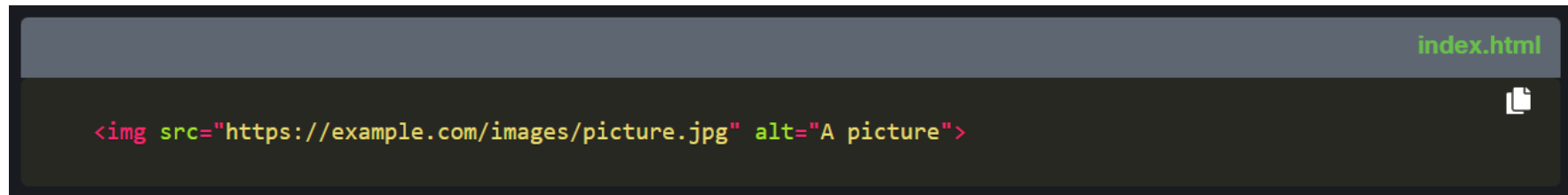
```

```

3.1.3 Image Location

Absolute Path

An absolute path is a path that starts from the root directory of the website. For example:



In this example, the absolute path points to an image file (picture.jpg) hosted on the web server at the URL "https://example.com/images/pic.jpg".

In addition to these methods, there are also several third-party image hosting services such as Imgur, Flickr, Photobucket, Google Photos, Dropbox, or Amazon S3, that allow you to store and manage your images online. These services can provide additional features such as automatic image resizing, optimization, and compression.

However, keep in mind that using third-party services may require an additional cost or subscription, and you may have limited control over the hosting and availability of your images.

3.1.4 Responsive Images: the srcset and sizes Attributes

As more and more people use mobile devices to browse the web, creating responsive images has become a crucial aspect of web design. Ensuring that images look good on screens of various sizes can greatly enhance the user experience and improve engagement.

To create responsive images in HTML, you can use the srcset and sizes attributes.

The srcset Attribute

The srcset attribute allows you to specify a list of image files with different resolutions, sizes, and formats. When the browser loads the image, it will select the most appropriate image from the list based on the screen size and the device's capabilities.

The sizes Attribute

The sizes attribute specifies the size of the image container in which the image will be displayed. In other words, it tells the browser how much screen space the image will occupy on different devices.

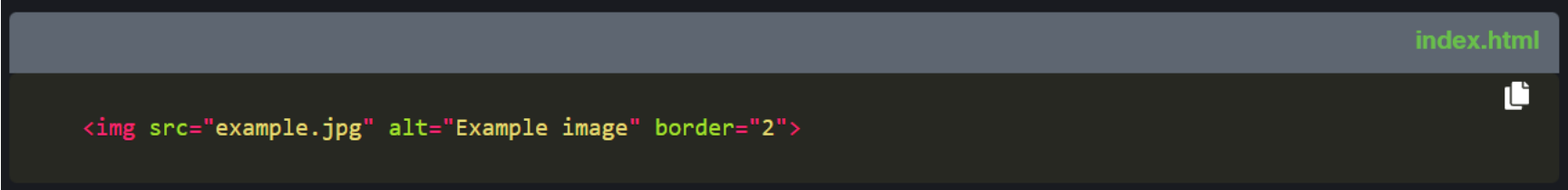
3.1.5 Other Image-Specific Attributes: border, hspace, vspace, and align

In addition to the commonly used src, alt, width, and height attributes for images in HTML, there are several other attributes that were historically used to modify an image's display. While these attributes offer a way to directly manipulate image presentation within the HTML markup, it's important to remember that modern web development practices favor CSS for such visual adjustments to ensure cleaner code and separation of content from presentation.

In this section, we will briefly explore these traditional attributes, acknowledging their role in the evolution of HTML while also emphasizing the shift towards CSS for achieving similar results.

The border Attribute

The border attribute specifies the width of the border around the image. For example:

A screenshot of a code editor interface. At the top right, a tab is labeled 'index.html' in green text. The main editing area has a dark background and shows the HTML code: ``. The code is color-coded: the opening tag is pink, the attribute names are green, and the values are yellow. A small icon of two overlapping sheets of paper is visible in the bottom right corner of the editor area.

```

```

3.1.5 Other Image-Specific Attributes: border, hspace, vspace, and align

The hspace and vspace Attributes

The hspace and vspace attributes specify the horizontal and vertical space around the image, respectively. For example:

```

```

The align Attribute

The align attribute allows for specifying the alignment of an image relative to surrounding content, with values like left, right, and center. For example:

```

```

3.1.6 Best Practices for Image Insertion in HTML

Things to keep in mind when working with images in HTML:

- Name your image files descriptively and include relevant keywords. This can help with search engine optimization and make it easier to find images in your image library. Avoid using generic names like "image1.jpg" or "photo.png".
- Use appropriate image formats for the type of image to ensure the best image quality and fast loading time.
- Store images in an appropriate location. Ensure that the image file is either in the same directory as your HTML file (relative path) or provide the full path to the image file in the src attribute (absolute path).
- Consider the file size of the image as large images can slow down your website's load time. It's a good idea to optimize images for the web before uploading them to your site (i.e. compress them to reduce file size without sacrificing image quality).
- Always provide descriptive and meaningful alt text for your images to improve accessibility and user experience.
- Last but not least, consider the placement of your images to complement your content and enhance the user experience. Avoid placing images in distracting locations that may interfere with the readability of

3.2 Hyperlinks

3.2.1 HTML Links

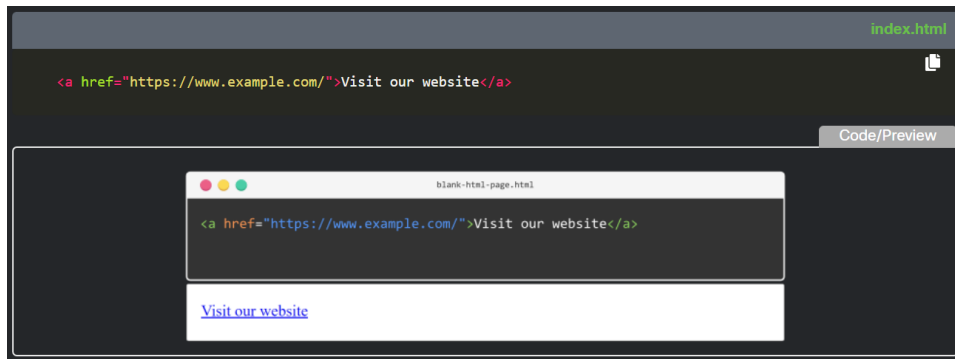
Hyperlinks (anchor links or simply links) are a fundamental part of the web. They are clickable elements that allow users to navigate between different pages on the same website, or to link to other websites and online resources.

In HTML, you can easily create links using the `<a>` tag, which stands for "anchor".

The `<a>` tag primarily relies on the `href` attribute, which specifies the destination of the link. This destination can be an absolute URL (such as `https://www.example.com/`) or a relative URL (like `about.html` if the target page resides in the same directory as the current page).

To add a visible, clickable link text in HTML, you need to place it between the opening `<a>` tag and the closing `</a>` tag. This text is what is displayed on the web page.

For example, the following code will create a hyperlink with the text "Visit our website" that points to the URL "https://www.example.com":



3.2.3 Internal Links

There are two types of hyperlinks: internal and external. Internal links are used to navigate within the same website or web page, while external links are used to navigate to other websites or web pages.

To create an internal link, you can use either an absolute or a relative URL to specify the URL of the page or section of the same website that you want to link to. For example:

- `<a href="https://example.com/page.html">Link to Page</a>` – the full address of the page (absolute URL).
- `<a href="/page.html">Link to Page</a>` – the relative URL which only includes the path to the page, and creates a link to the page.html file in the same directory as the current page.

3.2.4 Anchor Links

You can also create links that point to specific sections within a web page. Such links are sometimes called anchor links or jump links.

To create a link like this, you're going to need the id attribute and the # symbol. Here's how to do it:

1. Add the id attribute to an HTML element on the page that you want to link to. For example, if you want to link to an `<h2>` element with the content "Section 1", you can do the following: `<h2 id="section-1">Section 1</h2>`.
2. Then, create a link that points to this anchor by adding the # symbol followed by the id to the href attribute of an `<a>` tag, like this: `<a href="#section-1">Go to Section 1</a>`.

When a user clicks on this link, they will be taken to the section of the page with the `id="section-1"`. It's important to note that the id value must be unique within the page, and it's a good practice to use descriptive names that indicate the section's content.

3.2.5 External Links

To create an external link, you need to specify the full URL of the page, website, or resource that you want to link to. For example:

index.html



```
<a href="https://example.com">Visit Example.com</a>
```

It's important to note that because external links take users away from your website, you should ensure that they open in a new window or tab using the target attribute. For example:

index.html



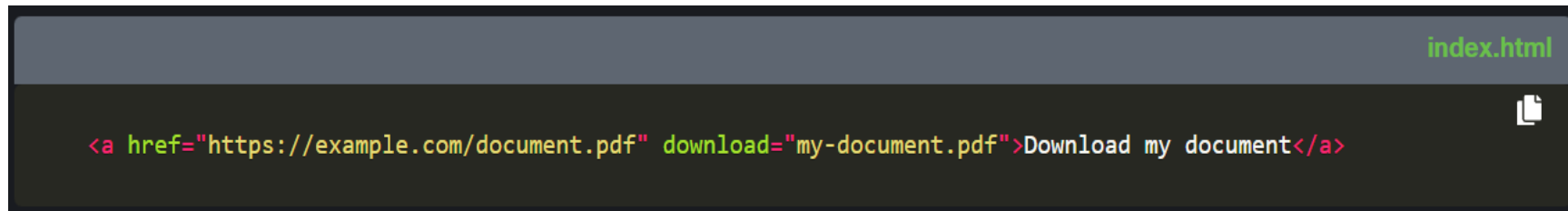
```
<a href="https://www.example.com" target="_blank">Visit Example Website</a>
```

3.2.7 Other Link-Specific Attributes: download, rel, and title

In addition to the commonly used href and target attributes, there are several other link-specific attributes that can be used to enhance the functionality and user experience of hyperlinks. Here are three examples: download, rel, and title.

The download Attribute

The download attribute specifies that the target of the link should be downloaded rather than navigated to. The attribute value can be used as the default filename that the downloaded file will be saved as.



3.2.7 Other Link-Specific Attributes: download, rel, and title

The rel Attribute

The rel attribute specifies the relationship between the current document and the target document. It can be used to indicate that the target document is a stylesheet, a parent document, or a part of a series, among other things.

index.html

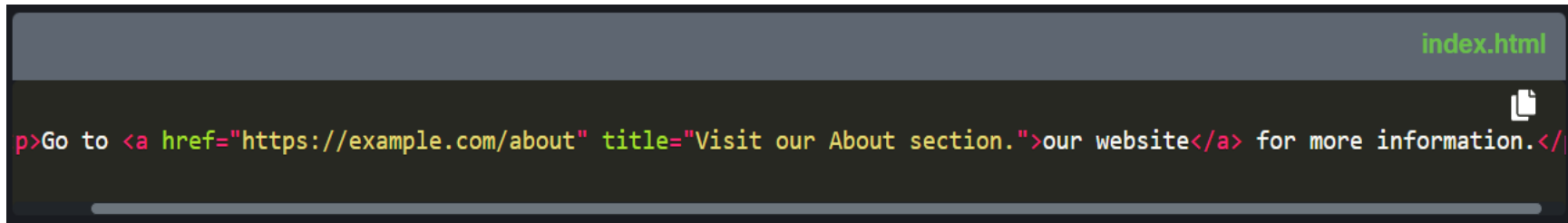
```
<a href="https://example.com/stylesheet.css" rel="stylesheet">Link to stylesheet</a>
```

3.2.7 Other Link-Specific Attributes: download, rel, and title

The title Attribute

The title attribute provides additional information about an element, enhancing the user's understanding and interaction with web content.

The attribute offers supplementary details about the hyperlink's destination or purpose. This information typically appears as a tooltip, a small pop-up box, when the user hovers their mouse over the link, giving users a preview of what to expect before clicking.



3.2.8 Turning Elements into Links

To make an element clickable, you simply need to wrap the `<a>` tag around it. The `href` attribute within this tag defines the URL to which the link points.

This method works with a wide range of elements, including images, blocks of text, and even complex HTML structures like `div`s or sections.

Here's a simple example that illustrates how you can turn an image into a clickable link:

```
<a href="https://www.example.com">  
    
</a>
```

In this example, the image becomes part of the clickable link. This is particularly useful for creating visually appealing call-to-action (CTA) buttons or linking card elements that contain a combination of text, images, and other media.

3.3 Media

3.3.1 Video and Audio Elements

Media content such as video and audio can greatly enhance a website's user experience. HTML provides the `<video>` and `<audio>` elements for embedding these types of media in a web page.

The elements support several attributes, such as:

- `src` for specifying the URL of the media file,
- `controls` for displaying built-in media controls,
- `width` and `height` for specifying the dimensions of the player,
- `autoplay` for automatically playing the media when the page loads,
- `loop` for looping playback, and
- `preload` for buffering the media file.

3.3.1 Video and Audio Elements

index.html

```
<video src="myvideo.mp4" controls autoplay>
  Your browser does not support the video tag.
</video>
```

Code/Preview

blank-html-page.html

```
<video src="myvideo.mp4" controls autoplay>
  Your browser does not support the video tag.
</video>
```

3.3.2 Ensuring Cross-Browser Compatibility

It's important to note that different browsers support different video and audio codecs, which can impact the compatibility of media content across various devices.

To ensure cross-browser compatibility, it's recommended to provide multiple formats of the same media file, such as MP4 and WebM for video, and MP3 and Ogg for audio. This can be achieved using the `<source>` element within the `<video>` and `<audio>` elements.

```
index.html

<!-- Video Example -->
<video controls>
  <source src="myvideo.mp4" type="video/mp4">
  <source src="myvideo.webm" type="video/webm">
  <p>Your browser does not support the video tag.</p>
</video>

<!-- Audio Example -->
<audio controls>
  <source src="myaudiofile.mp3" type="audio/mp3">
  <source src="myaudiofile.ogg" type="audio/ogg">
  <p>Your browser does not support the audio tag.</p>
</audio>
```

3.3.4 Responsive Videos

Just like with images, applying responsive design techniques to videos can ensure that your media content is accessible and engaging to users on any device, including desktop computers, tablets, and mobile phones.

To create a responsive video using HTML, you can use the `<video>` element with the `poster` and `controls` attributes, and the `<source>` element with the `media` attribute.

The `poster` attribute specifies an image that will be displayed when the video is not playing, while the `media` attribute located within the `<source>` element specifies a maximum width for the video to be displayed. This allows the video to adapt to different

index.html

```
<video poster="web-dev-picture.jpg" controls>
  <source src="web-dev-video.mp4" type="video/mp4" media="(max-width: 600px)">
  <source src="web-dev-video.webm" type="video/webm">
</video>
```